Illustrative examination

Original sample questions for demonstrating source mapping. These questions were not taken from a real past examination. Each question carries 5 illustrative marks.

Question 1 / 5 marks

Explain why two nested n-iteration loops have quadratic time complexity.

Question 2 / 5 marks

Compare arrays and linked lists for indexed access and insertion.

Question 3 / 5 marks

Explain the difference between a stack and a queue with applications.

Question 4 / 5 marks

Why is O(log n) search not guaranteed in every binary search tree?

Question 5 / 5 marks

Compare BFS and DFS, including their supporting structures.

Question 6 / 5 marks

Explain the divide-and-conquer structure of merge sort.